



Course name: INTERMEDIATE MACROECONOMICS
Course code: EC2211
Type of exam: Main
Examiner: Martin Flodén
Number of credits: 7.5
Date of exam: Tuesday 29 October 2024
Examination time: 5 hours (14:00-19:00)
Aids: No aids are allowed. You can only use pen, paper, eraser, ruler.

Write your anonymous code on each answer sheet (the code you received from Ladok when you registered for the exam).

Start each new question on a new answer sheet.

Explain notions/concepts and symbols. If you think that a question is vaguely formulated, specify the conditions used for solving it. Only legible exams will be marked.

The exam consists of 5 questions, worth 100 points in total. The maximum number of points on each sub-question are stated within parenthesis. For the grade E 45 points are required, for D 50 points, C 60 points, B 75 points and A 90 points.

All questions should be answered in English.

Your results will be made available on your Ladok account (www.student.ladok.se) within 15 working days from the date of the examination.

Good luck!



Question 1: Short questions (25 points in total)

Please write no more than one page (maximum) when answering each of these questions.

- a. Explain what is meant by Ricardian equivalence. Do you think that this theory is likely to hold up in reality? Why/why not? (5 points)
- b. Explain what is meant by purchasing power parity. Do you think that this theory is likely to hold up in reality? Why/why not? (5 points)
- c. State the Fisher equation. Explain what is meant by the Taylor principle and the logic behind that principle. (5 points)
- d. Consider an individual who lives for a single period, chooses how much to work and who has no other income than the wage income. She derives utility from consumption and disutility from working according to the following utility function:

$$U(C, H) = \ln(C) - \alpha H$$

Let w denote the wage rate. Formulate the individual's budget constraint and solve for the individual's optimal choices of consumption and hours worked. (5 points)

- e. Explain the difference between automatic stabilizers and active (discretionary) fiscal policy. Provide 3 examples of fiscal policy measures and explain to which of these two categories that they belong. Provide at least one example of each type. (5 points)



Question 2: The Solow model (25 points)

Consider a version of the Solow model where the population, L_t , grows at rate n while labor efficiency is constant and equal to unity. A fraction s of income is invested in capital every period and capital depreciates at rate δ . The production technology is Cobb-Douglas and given by:

$$Y_t = K_t^\alpha L_t^{1-\alpha}$$

where $0 < \alpha < 1$.

- Derive an expression for the accumulation of capital per person in this economy, i.e., Δk_{t+1} , where $k_t \equiv K_t/L_t$. (5 points)
- What is the steady state condition in this economy? Illustrate the equilibrium in a diagram. (4 points)
- Suppose that the economy starts out with a capital stock that is lower than k^* . Explain how the economy reaches the steady state, i.e., describe the mechanism. (4 points)
- Suppose now that the economy is in steady state. Analyze the effects of a permanent increase in the savings rate. Show how both output per person and consumption per person develop over time. Explain. (12 points)

Question 3: Aggregate supply and demand (20 points)

Consider an economy that begins with output at potential and an inflation rate at the target, $\pi_t = \pi^*$. A new chair of the central bank decides to raise the long-run inflation target to π^{**} (where $\pi^{**} > \pi^*$).

- Summarize the main assumptions behind the AS and AD curves. (4 points)
- Use the AS-AD model to show how the economy responds over time to the new inflation target. In answering this question, assume that the central bank cannot use communication to affect inflation expectations. (12 points)
- Suppose now that the central bank with the help of communication about the new inflation target *can* control inflation expectations. Would/could the transition to the new steady state then be different? Explain in words without using graphs or equations. (4 points)



Question 4: Economic growth (20 points)

This is an essay question where you are supposed to refer to theories and material covered in the course but you should not use equations or graphs in your answer. Write no more than 3 pages (maximum). Only legible answers will be considered.

Consider the Solow model, the Malthusian model and the Romer model. Explain in words the main assumptions, mechanisms and conclusions in these models. Focus in particular on elements where the models differ and make sure to describe how population growth relates to the development of income per capita in the three models.

Question 5: Growth accounting (10 points)

This is a credit question that should only be answered by students who have not received course credit from the seminar exercises. Students who have obtained course credit automatically receive 10 points on this question and cannot obtain extra points by answering it.

Consider an economy where the production function is

$$Y_t = A_t K_t^\alpha H_t^{1-\alpha}$$

where Y is output, A is TFP, K is capital and H is total hours worked in the economy. Assume that $\alpha = 1/3$.

Over the last 50 years, the average annual growth rates of output per capita, capital per capita and hours per capita have been $g_y = 2\%$, $g_k = 3\%$ and $g_h = 0.3\%$, respectively.

- a. Do the growth accounting for this economy by attributing growth in output per capita to the contributions from TFP, capital per capita and hours per capita, respectively. (4 points)
- b. Do the growth accounting for this economy by attributing growth in output per capita to the contributions from TFP, the capital-output ratio and hours per capita, respectively. (6 points)